

Short-Interest ETF Signal Strategy

A Concrete Research, Data, Mathematical Modeling, and
Backtest Execution Blueprint

Stock-first signal discovery → point-in-time ETF aggregation → allocation and risk control

Primary research thesis. ETFs whose constituents experience unusually large, newly reported increases in short interest should subsequently underperform ETFs whose constituents experience short covering or low short pressure. The first implementation is an **underweight/avoidance and cross-sectional ranking model**, not an automatic inverse-ETF strategy.

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Document Purpose

This document converts the short-interest concept into a reproducible quantitative research program. It specifies the data contracts, point-in-time rules, stock-level and ETF-level mathematical models, experiment sequence, portfolio variants, statistical tests, implementation architecture, and decision gates needed to determine whether the strategy contains genuine out-of-sample information.

The plan deliberately separates four questions:

1. Do newly disclosed changes in short interest predict future *stock* returns?
2. Does aggregating those changes through point-in-time ETF holdings improve the signal or reduce idiosyncratic squeeze risk?
3. Is the signal more valuable for avoiding/underweighting weak ETFs than for taking outright short or inverse positions?
4. Does short-interest information add value beyond price momentum, liquidity, size, sector effects, earnings events, and the other alpha engines in the broader trading system?

Important interpretation

A high short-interest value is not automatically a clean bearish signal. It can represent informed negative positioning, difficult-to-borrow securities, crowded consensus, hedging activity, capital-structure arbitrage, or fuel for a squeeze. The research therefore builds a **bearish-conviction engine** and a separate **squeeze-risk engine**. The two outputs must never be collapsed without explicit testing.

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Chapter 1

Executive Research Specification

1.1 Canonical research question

The primary question is:

Does a point-in-time, stock-first aggregation of newly reported short-interest changes identify ETFs that subsequently underperform their peers, after controlling for momentum, sector, liquidity, volatility, and earnings-related information?

The canonical stock signal is a sector-normalized change in short interest as a proportion of point-in-time float or shares outstanding:

$$B_{i,r}^{(1)} = Z_r^{\text{sector}}(\Delta s_{i,r}), \quad \Delta s_{i,r} = s_{i,r} - s_{i,r-1}, \quad (1.1)$$

where i denotes a stock and r denotes a public short-interest release cycle.

The initial ETF signal is:

$$S_{j,r}^{\text{ETF}} = \frac{\sum_i w_{ij,a_r} B_{i,r}^{(1)}}{\max\left(\sum_i w_{ij,a_r} \mathbf{1}[\text{signal observed for } i], \varepsilon\right)}, \quad (1.2)$$

where w_{ij,a_r} is the point-in-time weight of stock i in ETF j at the information-availability time a_r .

1.2 Primary implementation choice

The first monetization test is not “buy inverse ETFs.” It is:

1. rank eligible ETFs by $S_{j,r}^{\text{ETF}}$;
2. underweight or exclude the highest-short-pressure group;
3. overweight or hold the lowest-pressure/covering group;
4. measure whether this improves alpha, Sharpe ratio, drawdown, and downside capture after costs.

An inverse-ETF phase is permitted only if the signal predicts *negative absolute returns*, not merely relative underperformance.

1.3 Canonical first experiment

Frozen V1 specification

- Signal input: official-style reported short interest, not daily short-sale volume.
- Availability: public dissemination date/time, never settlement date.
- Stock score: sector-robust z-score of the change in short-interest ratio.
- Seed stocks: top 10% bearish scores and bottom 10% covering scores, tested separately.
- ETF discovery: point-in-time stock-to-ETF reverse index.
- ETF eligibility: U.S.-listed, unleveraged equity ETFs; minimum age, liquidity, mapping, and signal-coverage rules.
- ETF score: coverage-adjusted weighted stock score; breadth reported independently.
- Rebalance: next tradable market open after release.
- Primary holding rule: until the next public short-interest release; secondary horizons 5, 20, 40, and 60 trading days.
- Primary test: bottom-pressure quintile minus top-pressure quintile.
- Costs: 10 bps per side primary assumption, with 0/5/20 bps sensitivity.
- No borrow-cost model, ORTEX estimates, options, machine learning, or inverse ETFs in V1.

1.4 Expected strategic role

The most plausible role in the multi-engine architecture is:

$$\text{Short-interest signal} \approx \text{bearish capital-positioning and risk-allocation sensor} \quad (1.3)$$

It may provide return selection, but the lower hurdle is demonstrating that high-pressure ETFs deserve less capital or tighter risk budgets. This makes the signal naturally complementary to bullish analyst, insider-buying, earnings, and momentum engines.

Chapter 2

What the Data Actually Means

2.1 Short interest versus short-sale volume

Short interest is a snapshot of open short positions, while daily short-sale volume is transaction flow. FINRA explicitly cautions that the two datasets are not equivalent. U.S. firms currently report short-interest positions twice monthly under the existing regime.[1, 2]

Define:

$$SI_{i,r} = \text{shares sold short and still open for stock } i \text{ at settlement snapshot } r, \quad (2.1)$$

$$SSV_{i,d} = \text{shares executed as short-sale transactions on trading day } d. \quad (2.2)$$

In general,

$$SSV_{i,d} \neq \Delta SI_{i,r}, \quad (2.3)$$

because intraday short sales may be closed, offset, created for market-making, or otherwise absent from the later open-position snapshot.

Invalid V1 shortcut

Do not use daily short volume / total volume as a proxy for growth in open bearish positions. It can be studied later as a separate flow signal, but it must not be mislabeled as short interest.

2.2 Current reporting regime and regulatory note

As of August 22, 2026, FINRA's public reporting page still states that firms report short interest twice a month. A 2026 proposal to amend the frequency and granularity of Rule 4560 was withdrawn on August 5, 2026.[2, 3]

This matters operationally because the research clock is discrete. The strategy generally receives two official-style information updates per month, not a continuous daily series.

2.3 Settlement date is not availability date

For each record, preserve:

$$q_r = \text{position settlement/snapshot date}, \quad (2.4)$$

$$f_r = \text{regulatory filing deadline}, \quad (2.5)$$

$$a_r = \text{public dissemination timestamp}, \quad (2.6)$$

$$\tau_r = \text{first permitted execution timestamp}. \quad (2.7)$$

The point-in-time rule is:

$$\tau_r = \min\{u : u > a_r, u \text{ is a permitted market execution time}\}. \quad (2.8)$$

If only a release date is known, use the next regular-session open. If an exact pre-market timestamp is provably available and the trading model permits opening-auction orders, that can be a separate execution specification.

Look-ahead example

If the snapshot date is July 31 but the data becomes public on August 11, a July 31 or August 1 trade is invalid. The backtest may not update the signal until August 11, and the conservative primary entry is the next market open.

2.4 What official data omits

Reported short interest generally does not fully identify:

- the identity or type of the short seller;
- whether a position is directional, hedging, arbitrage, convertible-related, or market-making;
- synthetic short exposure created through derivatives;
- the complete economics of stock borrow, including fee, recall risk, and availability;
- intraperiod changes between the two snapshots.

These omissions motivate robustness tests and optional securities-lending enhancements, not ad hoc interpretation.

Chapter 3

Data Source and Collection Architecture

3.1 Recommended source stack

Layer	Recommended source		Use
Official reference	FINRA	and listing-exchange publications	Verify definitions, reporting cadence, release calendars, recent records, and selected vendor observations. FINRA public archives are not by themselves a complete long exchange-listed history for the desired research window.[4]
Historical feed	V1	Intrinio or another licensed vendor carrying official-style historical short interest	Obtain settlement date, current/previous short position, percentage change, average daily volume, and days-to-cover in normalized history. Intrinio documents these fields for its historical security endpoint.[5]
ETF constituents	QuantConnect	U.S. ETF Constituents	Point-in-time ETF constituents and weights. QuantConnect states coverage of roughly 2,650 U.S. ETF listings from June 2009, daily from 2015 and monthly before 2015, with possible publication delay up to one week.[7]
Prices and execution	QuantConnect/LEAN		Equity and ETF prices, corporate-action handling, order simulation, fees, slippage, and live/paper deployment architecture.
Fundamentals	PIT	fundamentals provider	Historical float or shares outstanding, market capitalization, sector, earnings dates, and security metadata. Never back-fill current float into history.
Optional V2 lending data	ORTEX	or institutional securities-lending feed	Cost-to-borrow, utilization, availability, and modeled short-interest estimates. ORTEX describes its short-interest and lending measures as proprietary/aggregated; keep them separate from official snapshots.[6]

3.2 Why the public FINRA files are not enough alone

FINRA's public data is valuable for validation, but its file archive notes that prior to June 2021 the downloadable data contains OTC positions only and does not reflect exchange-listed short interest.[4] A decade-plus exchange-listed backtest therefore normally requires a licensed historical provider, an exchange archive, or another verified institutional source.

3.3 Immutable data layers

Use three immutable layers:

```
data/  
  raw/  
    short_interest_vendor/  
    finra_validation/  
    etf_constituents/  
    prices/  
  normalized/  
    security_master/  
    short_interest_events/  
    etf_holdings/  
    release_calendar/  
  derived/  
    stock_short_features/  
    stock_seed_sets/  
    stock_to_etf_index/  
    etf_short_signals/  
  manifests/
```

Raw vendor responses are append-only. Parser changes create new normalized versions rather than rewriting raw history.

3.4 Collection manifest

Every snapshot must include:

```
{  
  "dataset_id": "short_interest_intrinio_20260822_v1",  
  "retrieved_at_utc": "2026-08-22T18:00:00Z",  
  "vendor": "intrinio",  
  "endpoint_version": "v2",  
  "start_settlement_date": "2013-01-01",  
  "end_settlement_date": "2026-08-15",  
  "record_count": 0,  
  "symbols_requested": 0,  
  "symbols_failed": 0,  
  "sha256": "...",  
  "collector_git_commit": "...",  
  "license_scope": "internal research"  
}
```

3.5 Canonical security master

Ticker is not a permanent identifier. The security master should include:

Field	Purpose
security_id	Internal immutable identifier.
ticker, valid-from, valid-to	Time-bounded ticker mapping.
qc_symbol_id	QuantConnect mapping/security-master identifier where available.
vendor_security_id	Stable vendor identifier used for collection.
share_class, primary flag	Prevent accidental aggregation of distinct share classes.
exchange, country, security type	Universe and eligibility filters.
sector, industry, valid dates	Point-in-time normalization groups.
delisting_date, corporate-action links	Survivorship and return handling.

3.6 Normalized short-interest event schema

Field	Definition
event_id	Hash of vendor, security, settlement date, and version.
security_id	Canonical stock identifier.
settlement_date	Date of the position snapshot.
release_timestamp	First public availability timestamp used by the backtest.
current_short_shares	Current reported open short position.
previous_short_shares	Prior reporting-cycle position.
avg_daily_volume	Vendor/exchange average volume definition, retained exactly.
days_to_cover	Vendor field and independently recomputed version.
float_shares_pit	Point-in-time free float, with source and observation timestamp.
shares_outstanding_pit	Fallback denominator when float is unavailable.
source, source_record_id	Lineage and audit.
raw_payload_hash	Detect revisions or vendor restatements.

3.7 Availability calendar construction

Build a release calendar keyed by settlement date and listing venue. Use the actual public dissemination field when supplied. Otherwise map settlement dates to verified exchange publication dates and assign a conservative release time. The algorithm must never infer availability simply from the vendor file's retrieval date.

Chapter 4

Stock-Level Mathematical Models

4.1 Core raw measures

For stock i and release cycle r :

$$s_{i,r}^F = \frac{SI_{i,r}}{Float_{i,r}}, \quad \text{short interest as fraction of float,} \quad (4.1)$$

$$s_{i,r}^O = \frac{SI_{i,r}}{SharesOutstanding_{i,r}}, \quad \text{fallback fraction of shares outstanding,} \quad (4.2)$$

$$DTC_{i,r} = \frac{SI_{i,r}}{ADV_{i,r}^{(K)}}, \quad \text{days to cover,} \quad (4.3)$$

$$\Delta s_{i,r} = s_{i,r} - s_{i,r-1}, \quad (4.4)$$

$$g_{i,r} = \log\left(\frac{s_{i,r} + \varepsilon}{s_{i,r-1} + \varepsilon}\right), \quad (4.5)$$

$$a_{i,r} = \Delta s_{i,r} - \Delta s_{i,r-1}, \quad (4.6)$$

$$\Delta DTC_{i,r} = DTC_{i,r} - DTC_{i,r-1}. \quad (4.7)$$

Use s^F as primary only when the float series is genuinely point-in-time. Otherwise freeze s^O as the canonical denominator and test float-based results as a separate specification.

4.2 Robust sector normalization

Shorting norms differ by sector, capitalization, and security type. Use a robust cross-sectional z-score:

$$Z_{i,r}^{\text{sector}}(x) = \frac{x_{i,r} - \text{median}_{k \in \mathcal{S}(i,r)} x_{k,r}}{1.4826 \text{ MAD}_{k \in \mathcal{S}(i,r)}(x_{k,r}) + \varepsilon}, \quad (4.8)$$

where $\mathcal{S}(i, r)$ is the point-in-time sector peer set.

Winsorize inputs using a pre-registered rule, such as:

$$\tilde{x}_{i,r} = \text{clip}(x_{i,r}, Q_{0.01,r}(x), Q_{0.99,r}(x)). \quad (4.9)$$

Do not choose winsorization thresholds after inspecting strategy returns.

4.3 Stock-specific historical surprise

A 3 percentage-point increase is more unusual for a company that normally changes by 0.2 points than for one that routinely changes by 5 points. Define:

$$Z_{i,r}^{\text{hist}}(\Delta s) = \frac{\Delta s_{i,r} - \text{median}_{u \in [r-L, r-1]} \Delta s_{i,u}}{1.4826 \text{ MAD}_{u \in [r-L, r-1]}(\Delta s_{i,u}) + \varepsilon}. \quad (4.10)$$

Primary lookback candidates are $L \in \{12, 24\}$ release cycles, treated as separate pre-registered tests.

4.4 Pre-registered stock models

4.4.1 Model S0: short-interest level

$$B_{i,r}^{(0)} = Z_{i,r}^{\text{sector}}(s_{i,r}). \quad (4.11)$$

This is a baseline, not the preferred model.

4.4.2 Model S1: new short pressure - canonical

$$B_{i,r}^{(1)} = Z_{i,r}^{\text{sector}}(\Delta s_{i,r}) \quad (4.12)$$

A higher value means unusually strong new short positioning relative to sector peers.

4.4.3 Model S2: sector and historical surprise

$$B_{i,r}^{(2)} = \frac{1}{2}Z_{i,r}^{\text{sector}}(\Delta s) + \frac{1}{2}Z_{i,r}^{\text{hist}}(\Delta s). \quad (4.13)$$

Equal weighting is fixed because both components are standardized. Alternative weights require a separate experiment.

4.4.4 Model S3: pressure plus cover difficulty

$$B_{i,r}^{(3)} = \frac{1}{2}Z_{i,r}^{\text{sector}}(\Delta s) + \frac{1}{2}Z_{i,r}^{\text{sector}}(\Delta DTC). \quad (4.14)$$

This asks whether new short positioning is more informative when aggregate exit capacity is deteriorating.

4.4.5 Model S4: residual short-interest surprise

Estimate only from prior data:

$$\Delta s_{i,r} = \beta_0 + \beta_1 s_{i,r-1} + \beta_2 \log(MC_{i,r}) + \beta_3 Vol_{i,r} + \beta_4 Mom_{i,r} + \gamma_{\text{sector}} + u_{i,r}. \quad (4.15)$$

Then define:

$$B_{i,r}^{(4)} = Z_{i,r}^{\text{sector}}(u_{i,r}). \quad (4.16)$$

The residual is the portion of short-interest change not explained by common observable characteristics. This is a later round, not V1.

4.5 Price confirmation is a separate dimension

Let:

$$M_{i,r}^{(20)} = \frac{P_{i,a_r-1}}{P_{i,a_r-21}} - 1. \quad (4.17)$$

Classify, rather than immediately blend:

$$\text{Bearish confirmation : } B_{i,r} > q_{0.9}(B), \quad M_{i,r}^{(20)} < 0, \quad (4.18)$$

$$\text{Crowded disagreement : } B_{i,r} > q_{0.9}(B), \quad M_{i,r}^{(20)} > 0. \quad (4.19)$$

The first may represent informed bearish positioning; the second may represent unresolved disagreement or squeeze risk. Both must be reported separately.

4.6 Short-covering bullish signal

Define:

$$C_{i,r} = -B_{i,r}^{(1)}. \quad (4.20)$$

Large $C_{i,r}$ indicates unusually rapid short covering. Do not assume symmetric return effects. Test the covering portfolio independently because prior research finds that low or declining short interest can sometimes contain stronger positive information than the negative effect of high short interest.^[12]

4.7 Squeeze-risk engine

The squeeze engine is a risk classifier, not the bearish alpha score.

A V1-lite indicator is:

$$Q_{i,r}^{\text{lite}} = \mathbf{1}[s_{i,r} > Q_{0.90}(s)] \mathbf{1}[DTC_{i,r} > Q_{0.80}(DTC)] \mathbf{1}[M_{i,r}^{(20)} > 0]. \quad (4.21)$$

With lending data, a continuous score can be tested:

$$Q_{i,r} = \text{sigmoid}\left(\theta_1 Z(s) + \theta_2 Z(DTC) + \theta_3 Z(CTB) + \theta_4 Z(Util) + \theta_5 Z(M^{(20)}) - \theta_6 Z(Float)\right). \quad (4.22)$$

The θ values must be trained only within walk-forward training windows or replaced by a pre-registered equal-weight score. Never optimize them on the final holdout.

Default squeeze rule

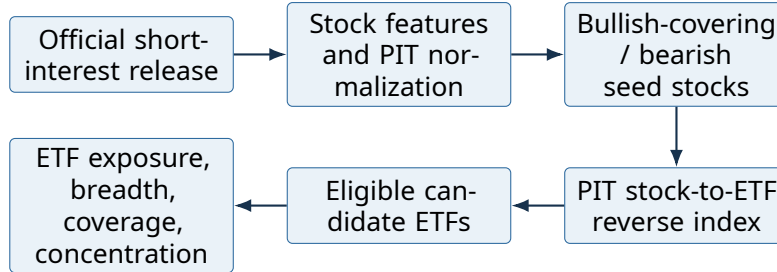
Exclude $Q^{\text{lite}} = 1$ stocks from direct single-stock short tests. For ETF aggregation, retain them in diagnostic calculations but cap or separately report their contribution. The goal is to learn whether ETF diversification absorbs squeeze risk, not to assume it does.

Chapter 5

Stock-First Seeds and ETF Discovery

5.1 Why stock-first is the correct architecture

The economic event occurs at the stock level. The strategy should therefore begin with newly observed short-pressure stocks and then discover the ETFs that concentrate those stocks:



This design discovers sector or thematic clusters from the data instead of forcing a predefined sector view.

5.2 Seed definitions

Let $p_{i,r}$ be the cross-sectional percentile rank of $B_{i,r}$ among eligible stocks.

Primary bearish seeds:

$$\mathcal{I}_r^- = \{i : p_{i,r} \geq 0.90\}. \quad (5.1)$$

Primary covering seeds:

$$\mathcal{I}_r^+ = \{i : p_{i,r} \leq 0.10\}. \quad (5.2)$$

Secondary thresholds 5%, 15%, and 20% are separate experiments. Do not tune the threshold on the final holdout.

5.3 Minimum data eligibility for stocks

A stock is eligible at release r only if:

- the security existed and was tradable at a_r ;
- both current and prior short-interest observations are valid;
- the denominator is point-in-time and positive;
- price and volume history meet the chosen lookback requirement;
- sector/industry mapping is available at a_r ;

- no unresolved split, ticker, merger, or share-class mapping error exists;
- market capitalization and liquidity exceed pre-registered minima for the stock-level validation sample.

5.4 Point-in-time reverse index

Define:

$$\mathcal{E}_{i,t} = \{(j, w_{ij,t}) : i \text{ is held by ETF } j \text{ at } t\}. \quad (5.3)$$

The candidate ETF set is:

$$\mathcal{J}_r = \bigcup_{i \in \mathcal{I}_r^- \cup \mathcal{I}_r^+} \{j : (j, w_{ij,a_r}) \in \mathcal{E}_{i,a_r}\}. \quad (5.4)$$

This reverse index should be precomputed by date from historical ETF holdings. A live algorithm may read date-partitioned reverse-index files rather than scanning thousands of ETF holdings at every event.

5.5 ETF eligibility

The alpha-discovery universe initially excludes leveraged, inverse, bond, commodity, volatility, crypto, and closed-end products. A candidate ETF must satisfy, point-in-time:

- U.S.-listed equity ETF classification;
- age of at least 252 trading days;
- median daily dollar volume above a frozen threshold, initially \$5 million;
- price above \$5;
- at least 80% of reported holding weight mapped to canonical security IDs;
- minimum observed short-signal coverage, initially 50%;
- no unresolved liquidation/merger event;
- no use of current AUM or current holdings to determine historical eligibility.

If point-in-time AUM is available, test an additional AUM threshold. Do not use current AUM as a historical filter.

Chapter 6

ETF-Level Mathematical Models

6.1 Observed coverage

For ETF j at release r :

$$Cov_{j,r} = \sum_i w_{ij,a_r} \mathbf{1}[B_{i,r} \text{ is observed}]. \quad (6.1)$$

Coverage is a first-class quality variable. An ETF with 20% signal coverage cannot be treated as equivalent to one with 90% coverage.

6.2 Raw weighted short-pressure exposure

$$E_{j,r} = \sum_i w_{ij,a_r} B_{i,r}. \quad (6.2)$$

This assigns more influence to a 10% holding than a 0.2% holding.

6.3 Coverage-adjusted exposure - canonical ETF score

$$\bar{E}_{j,r} = \frac{E_{j,r}}{\max(Cov_{j,r}, \varepsilon)} \quad (6.3)$$

This is the mean signal per observed dollar of ETF weight. The raw and adjusted scores must both be reported; adjustment can amplify noise when coverage is near the threshold.

6.4 Weighted breadth

For a bearish seed threshold c_r :

$$Br_{j,r}^{w,-} = \sum_i w_{ij,a_r} \mathbf{1}[B_{i,r} \geq c_r]. \quad (6.4)$$

Similarly, covering breadth is:

$$Br_{j,r}^{w,+} = \sum_i w_{ij,a_r} \mathbf{1}[B_{i,r} \leq -c_r]. \quad (6.5)$$

6.5 Count breadth

$$Br_{j,r}^{n,-} = \frac{\sum_i \mathbf{1}[w_{ij,a_r} > 0] \mathbf{1}[B_{i,r} \geq c_r]}{\sum_i \mathbf{1}[w_{ij,a_r} > 0] \mathbf{1}[B_{i,r} \text{ observed}]}. \quad (6.6)$$

Weighted breadth measures affected capital exposure; count breadth measures how broadly the signal is distributed among observed holdings.

6.6 Signal concentration and effective seed count

Let the positive bearish contribution of stock i be:

$$c_{ij,r} = w_{ij,a,r} \max(B_{i,r}, 0). \quad (6.7)$$

Normalize:

$$\pi_{ij,r} = \frac{c_{ij,r}}{\sum_k c_{kj,r} + \varepsilon}. \quad (6.8)$$

Then:

$$HHI_{j,r}^{sig} = \sum_i \pi_{ij,r}^2, \quad (6.9)$$

$$N_{j,r}^{eff} = \frac{1}{HHI_{j,r}^{sig} + \varepsilon}. \quad (6.10)$$

A high N^{eff} means the ETF signal is supported by multiple independent constituent contributions. This may be safer than an ETF score dominated by one squeeze-prone stock.

6.7 Candidate ETF model battery

Model	Formula	Question
E0	$\sum_i w_{ij} Z(s_i)$	Does absolute short-interest level aggregate to ETFs?
E1	$\bar{E}_{j,r}$ using $B^{(1)}$	Does newly reported short pressure predict ETF underperformance?
E2	$Br_{j,r}^{w,-}$	Does broad affected portfolio weight matter more than magnitude?
E3	$Br_{j,r}^{n,-}$	Does the number of affected constituents matter?
E4	$(Z(\bar{E}) + Z(Br^{w,-}))/2$	Does standardized exposure plus breadth improve OOS results?
E5	$\bar{E} \sqrt{Br^{w,-}} \sqrt{1 - HHI^{sig}}$	Does broad, diversified bearish pressure outperform concentrated pressure?
E6	covering counterpart	Does short covering identify stronger ETFs?

E1, E2, and E3 must be tested independently before E4/E5. A complex composite is not automatically superior.

6.8 ETF overlap control

Many ETFs represent nearly the same holdings. Define holdings cosine similarity:

$$Sim_{ab,r} = \frac{\sum_i w_{ia,r} w_{ib,r}}{\sqrt{\sum_i w_{ia,r}^2} \sqrt{\sum_i w_{ib,r}^2}}. \quad (6.11)$$

Cluster ETFs with $Sim_{ab,r} \geq 0.90$ in the primary sensitivity test. Run:

1. all eligible ETFs;
2. one representative ETF per holdings cluster, chosen by point-in-time liquidity;
3. cluster-equal-weight observations;
4. standard errors clustered by release date and ETF cluster.

This prevents a semiconductor theme represented by several near-duplicate funds from masquerading as several independent discoveries.

Chapter 7

Trading and Allocation Variants

7.1 Variant A: underweight/avoidance - primary

Let $w_{j,r}^{base}$ be the baseline allocator's weight. Remove the top-pressure decile:

$$w_{j,r}^{avoid} = \frac{w_{j,r}^{base} \mathbf{1}[S_{j,r}^{ETF} < Q_{0.90,r}(S)]}{\sum_k w_{k,r}^{base} \mathbf{1}[S_{k,r}^{ETF} < Q_{0.90,r}(S)]}. \quad (7.1)$$

Compare the filtered portfolio with the same baseline universe and rebalance schedule.

This directly tests whether the signal improves risk selection without requiring negative absolute returns.

7.2 Variant B: long low-pressure/covering ETFs

Hold the lowest-pressure quintile or the strongest-covering quintile:

$$w_{j,r}^{low} = \frac{\mathbf{1}[j \in Q_{low,r}]/\hat{\sigma}_{j,r}}{\sum_{k \in Q_{low,r}} 1/\hat{\sigma}_{k,r}}, \quad (7.2)$$

where inverse-volatility weighting is a secondary specification; equal weighting is primary.

7.3 Variant C: long-short spread

$$R_r^{LS} = R(Q_{low,r}) - R(Q_{high,r}). \quad (7.3)$$

Test:

- equal-dollar long-short;
- beta-neutral long-short;
- sector-neutral long-short;
- one representative ETF per overlap cluster.

The long-short spread is primarily a research diagnostic. Practical shorting costs and borrow availability of ETFs must be modeled before treating it as executable.

7.4 Variant D: inverse ETF overlay - gated

Only after validation, map an ordinary ETF j to a historical inverse product j^{-1} that existed at the trade date. A conditional rule may be:

$$TradeInverse_{j,r} = \mathbf{1}[S_{j,r}^{ETF} \geq Q_{0.95}(S)] \mathbf{1}[M_{20,j,r}^{ETF} < 0] \mathbf{1}[M_{200,r}^{SPY} < 0]. \quad (7.4)$$

The first inverse phase uses only 1x inverse products and short holding windows. Daily-reset leveraged inverse products require path-dependent return modeling and a separate experiment.

Inverse ETF admission gate

Proceed only if the top-pressure group has negative absolute net returns across multiple out-of-sample periods and regimes. Relative underperformance alone is not sufficient.

7.5 Variant E: risk-budget input

Rather than placing a trade, feed the standardized ETF score to the central risk engine:

$$M_{j,r}^{\text{risk}} = \exp[-\kappa \max\{0, Z(S_{j,r}^{\text{ETF}})\}]. \quad (7.5)$$

The primary non-parametric version uses discrete multipliers by quintile. The continuous κ model is later tuned only in training windows.

7.6 Keep squeeze trading separate

Potential squeeze candidates should be researched as a distinct long/tactical strategy. They should not be treated as “less bearish” merely because their prices are rising. Maintain separate experiment IDs, data features, and portfolio accounting.

Chapter 8

Backtest Execution Protocol

8.1 Event clock

The natural rebalance clock is the public short-interest release cycle. For each release r :

1. load all observations with $release_timestamp \leq a_r$;
2. build point-in-time stock features;
3. select stock seeds;
4. query the date-valid reverse ETF index;
5. apply ETF eligibility and coverage rules;
6. compute ETF scores;
7. submit orders at τ_r , the next permitted execution time;
8. hold until the next release for the primary strategy, or until a fixed horizon in secondary event studies.

8.2 As-of joins

For any field x :

$$x_{i,t} = x_{i,u^*}, \quad u^* = \max\{u : available_time(x_{i,u}) \leq t\}. \quad (8.1)$$

Every join must use availability time, not just effective date.

8.3 Primary forward-return definitions

For diagnostics, define open-to-open return after the conservative entry:

$$R_{j,r}^{(h)} = \frac{P_{j,\tau_r+h}^{open}}{P_{j,\tau_r}^{open}} - 1. \quad (8.2)$$

The portfolio engine should separately compute actual total-return P&L with dividends, splits, fees, and slippage.

Excess return is:

$$AR_{j,r}^{(h)} = R_{j,r}^{(h)} - R_{b(j),r}^{(h)}, \quad (8.3)$$

where $b(j)$ is SPY for market-adjusted results and a point-in-time sector benchmark for sector-adjusted results.

8.4 Primary horizons

- Natural holding period: from one public release to the next.
- Fixed primary horizon: 20 trading days.
- Secondary horizons: 5, 40, 60, and 120 trading days.
- One-day results: diagnostic only because the official signal is low frequency and dissemination timing may vary.

8.5 Cross-sectional sort test

At each release, sort eligible ETFs into quintiles:

Q_1 = highest short pressure, $\dots$, Q_5 = lowest pressure/strongest covering.

The expected relation is:

$$R(Q_5) > R(Q_4) > R(Q_3) > R(Q_2) > R(Q_1). \quad (8.4)$$

Primary spread:

$$Spread^{(h)} = R^{(h)}(Q_5) - R^{(h)}(Q_1). \quad (8.5)$$

Report equal-weight, inverse-volatility, and overlap-cluster-adjusted versions, with equal weight as primary.

8.6 Information coefficient

For each release:

$$IC_r^{(h)} = Corr_{Spearman} \left(-S_{j,r}^{ETF}, R_{j,r}^{(h)} \right). \quad (8.6)$$

The minus sign makes positive IC indicate that lower short pressure predicts higher returns. Report mean IC, median IC, ICIR, t-statistic, and percentage of positive periods.

8.7 Fama-MacBeth style cross-sectional regression

At each release:

$$AR_{j,r}^{(h)} = \alpha_r + \beta_r S_{j,r}^{ETF} + \delta_r' X_{j,r} + \epsilon_{j,r}. \quad (8.7)$$

Then test:

$$\bar{\beta} = \frac{1}{T} \sum_r \beta_r < 0. \quad (8.8)$$

Controls X include ETF momentum, realized volatility, size/liquidity, sector, concentration, and coverage. Use Newey-West or block-bootstrap inference appropriate for overlapping horizons.

8.8 Stock-level validation before ETF aggregation

Before interpreting ETF results, repeat quintile and regression tests at the stock level. Required comparisons:

- stock signal only;
- ETF signal using weighted aggregation;
- ETF simple seed count;
- conventional sector-average short signal;
- random ETF basket placebo.

The research is strongest if ETF aggregation improves stability or Sharpe relative to direct stock exposure while preserving predictive information.

8.9 Transaction cost model

Primary ETF assumptions:

$$Cost_{j,t} = Commission_{j,t} + \frac{1}{2} Spread_{j,t} + Impact_{j,t}, \quad (8.9)$$

$$Impact_{j,t} = \eta \sigma_{j,t} \sqrt{\frac{|TradeDollar_{j,t}|}{ADV_{j,t}^{\$}}}. \quad (8.10)$$

For the first battery, use deterministic cost scenarios of 0, 5, 10, and 20 bps per side. Later replace these with quote-based spreads and calibrated market impact.

For short/inverse phases, also model:

- ETF borrow fee and hard-to-borrow failures for direct shorts;
- inverse ETF expense ratios and tracking effects;
- daily reset/path dependency for leveraged products;
- slippage under stressed volatility.

8.10 Overlapping observations and standard errors

Twenty- and sixty-day forward returns overlap across twice-monthly releases. Naive t-statistics are invalid. Use:

- Newey-West lags at least proportional to the forward horizon;
- release-cycle block bootstrap;
- clustering by release date and ETF-overlap cluster;
- non-overlapping subsample sensitivity tests.

Chapter 9

Pre-Registered Experiment Sequence

9.1 Round SI-0: data integrity

No alpha results are allowed until SI-0 passes.

Required tests:

```
test_release_timestamp_after_or_equal_settlement_date()
test_signal_not_available_before_release()
test_float_is_point_in_time()
test_previous_short_position_matches_prior_cycle()
test_split_adjustments_consistent()
test_ticker_mapping_point_in_time()
test_delisted_securities_retained()
test_etf_holdings_not_future_dated()
test_etf_weight_sum_reasonable()
test_mapping_weight_at_least_80_percent()
test_no_duplicate_security_release_records()
```

Output disposition: PASS, FAIL, or INVALID-DATA.

9.2 Round SI-1: stock-level signal

Test S0, S1, and short covering. No ETF aggregation.

Primary hypothesis:

$$\mathbb{E}[R^{(20)} \mid B^{(1)} \text{ high}] < \mathbb{E}[R^{(20)} \mid B^{(1)} \text{ low}]. \quad (9.1)$$

Required outputs include deciles, IC, regression coefficients, and sector splits. Also report year-by-year results and squeeze-flag splits.

9.3 Round SI-2: simple ETF aggregation

Use E1, point-in-time weights, 50% coverage, next-open execution, and 20-day/natural-cycle holding periods. Compare against simple hit count.

9.4 Round SI-3: breadth and concentration

Test E2, E3, E4, E5 and N^{eff} . Freeze formulas before results.

9.5 Round SI-4: confounders

Control for:

- 1-, 3-, 6-, and 12-month stock/ETF momentum;
- 20- and 60-day realized volatility;
- market capitalization, float, and liquidity;
- sector and industry fixed effects;
- earnings-announcement windows;
- analyst revisions and insider purchases, if those engines are available;
- valuation/fundamental deterioration;
- high-squeeze flags.

9.6 Round SI-5: realistic long-only allocation

After costs, test the underweight/avoidance, low-pressure long, and risk-budget variants. Evaluate the combined portfolio, not only a standalone ETF sleeve.

9.7 Round SI-6: bearish implementation

Only after SI-5 passes, compare:

1. no trade / avoidance;
2. underweight ordinary ETF;
3. short ordinary ETF;
4. 1x inverse ETF;
5. 2x inverse ETF, last and separately.

9.8 Round SI-7: multi-engine integration

Test incremental value conditional on analyst, insider, earnings, momentum, and regime signals. Do not combine raw scores without out-of-sample normalization.

Chapter 10

Confounders, Robustness, and Placebos

10.1 Momentum

Short sellers may enter after price run-ups, while price declines may validate bearish information. Test:

$$AR_{j,r+h} = \alpha + \beta_1 S_{j,r}^{ETF} + \beta_2 Mom_{j,r} + \beta_3 Vol_{j,r} + \epsilon. \quad (10.1)$$

The short-interest coefficient must retain incremental economic value after momentum controls.

10.2 Earnings windows

Run:

- all observations;
- exclude stocks with earnings within ± 2 trading days of the snapshot or release;
- exclude within ± 5 trading days;
- separate pre-earnings from post-earnings changes.

If the signal disappears, it may be an earnings-information proxy rather than a distinct channel.

10.3 Liquidity and small-cap concentration

High short interest is often concentrated in smaller, less liquid stocks. Require:

- stock-level size/liquidity controls;
- ETF-level liquidity filters;
- value-weighted and equal-weighted results;
- large-cap-only and small-cap-excluded sensitivity tests.

Prior research is not uniformly strong across weighting schemes. Therefore, report both economic and statistical robustness instead of relying on one portfolio sort.[\[10\]](#)

10.4 Squeeze and crowding splits

Report high-pressure stocks/ETFs by:

- positive versus negative recent momentum;
- high versus low days-to-cover;

- high versus low float;
- high versus low borrow cost/utilization when available;
- options-implied squeeze indicators, only in later research.

10.5 Corporate actions and structural positions

Flag:

- convertible issuance and merger arbitrage;
- SPACs and de-SPACs;
- ADRs and multiple share classes;
- recent IPOs and lock-up expirations;
- tender offers, rights offerings, and bankruptcies;
- extreme stock splits and reverse splits.

Run results with and without these categories.

10.6 Negative controls

1. Shuffle short-interest changes across stocks within sector/date.
2. Shuffle release dates within calendar year.
3. Randomly permute stock-to-ETF mappings within sector.
4. Use current ETF holdings as an intentionally invalid placebo; it should trigger the PIT audit, not become a reported strategy.
5. Replace the true signal with random ranks preserving the number of seeds.
6. Shift the signal one release backward and forward; the forward shift must never be used in a valid strategy and should expose leakage sensitivity.

10.7 Multiple testing

Label all specifications as PRIMARY, SECONDARY, or EXPLORATORY. Apply false-discovery-rate control to large batteries. The final investment conclusion must be based on frozen primary tests and untouched out-of-sample periods.

Chapter 11

Walk-Forward Validation Gates

11.1 Walk-forward design

Use expanding or rolling windows. Example:

Window	Train	Validate	Test
1	2013–2017	2018–2019	2020
2	2013–2018	2019–2020	2021
3	2013–2019	2020–2021	2022
4	2013–2020	2021–2022	2023
5	2013–2021	2022–2023	2024

Dates must be adjusted to actual data coverage. Final performance is the concatenation of test windows only.

11.2 Untouched holdout

Reserve a final period that has not been inspected during design. If 2024–2026 is used during exploratory work, it is no longer a valid holdout. Record every viewed period in the research log.

11.3 Signal-validation gate

Proceed from SI-2 to portfolio construction only if most of the following hold:

GO criteria for ETF signal

- All PIT, survivorship, and mapping audits pass.
- Mean OOS IC has the expected sign and is stable enough to be economically interpretable.
- $Q_5 - Q_1$ is positive after 10 bps per side.
- Quintile/decile returns are approximately monotonic.
- The result appears in at least 60% of OOS years or release-year blocks.
- Momentum and liquidity controls do not eliminate the effect.
- No single year, sector, or near-duplicate ETF cluster explains most of the result.
- Breadth/concentration diagnostics support an interpretable mechanism.

11.4 Risk-management gate

The underweight model may pass even if outright shorting fails. Accept as a risk signal if it materially improves at least two of:

- Sharpe or Sortino ratio;
- maximum drawdown;
- expected shortfall;
- downside capture;
- sector concentration;
- time underwater;

without an unacceptable loss of net CAGR.

11.5 Inverse implementation gate

NO-GO conditions

Do not advance to inverse ETFs if the high-pressure group merely rises less than peers, if performance is driven by one bear market, if costs erase the edge, if squeeze regimes dominate losses, or if negative absolute returns fail to persist across OOS periods.

Chapter 12

QuantConnect/LEAN Implementation

12.1 Platform role

QuantConnect can supply historical prices and order simulation. It also provides the U.S. ETF Constituents universe.

Its documentation supports both historical constituent queries and custom external datasets.[\[7, 8\]](#)

The short-interest feed should be preprocessed into immutable, chronological files and loaded as custom data. Do not call a mutable vendor API from inside a historical backtest.

12.2 Recommended artifact flow

```
Vendor download
-> raw JSON/CSV snapshot
-> normalize identifiers and release timestamps
-> calculate PIT stock features
-> build date-partitioned stock-to-ETF reverse index
-> upload immutable files to Object Store
-> LEAN custom data reader
-> event-driven ETF scoring and orders
```

12.3 Custom data row

```
class ShortInterestEvent(PythonData):
    # Time is PUBLIC availability, not settlement date.
    security_id: str
    settlement_date: datetime
    short_shares: float
    short_ratio: float
    delta_short_ratio: float
    sector_z_delta: float
    days_to_cover: float
    squeeze_flag: int
```

12.4 Algorithm event loop pseudocode

```
def on_short_interest_release(release_time, stock_events):
    assert all(e.time <= release_time for e in stock_events)

    stock_features.update(stock_events)
```

```

bearish_seeds = select_top_percentile(stock_features, 0.90)
covering_seeds = select_bottom_percentile(stock_features, 0.10)

candidate_etfs = set()
for stock in bearish_seeds | covering_seeds:
    candidate_etfs |= reverse_index.lookup(
        as_of=release_time,
        stock_id=stock.security_id
    )

scored = []
for etf in candidate_etfs:
    holdings = holdings_store.as_of(etf, release_time)
    if not etf_is_eligible(etf, holdings, release_time):
        continue
    score = calculate_etf_short_metrics(holdings, stock_features)
    scored.append(score)

target_weights = allocator.build_underweight_portfolio(scored)
schedule_orders_for_next_market_open(target_weights)

```

12.5 Precomputed reverse-index schema

```

as_of_date,stock_security_id,etf_symbol,etf_security_id,weight,
holdings_available_time,source_version
2026-08-20,SEC123,SMH,ETF009,0.0831,
2026-08-20T00:00:00Z,qc_etf_v1

```

12.6 Configuration file

```

experiment_id: SI-R2-E1-V1
classification: PRIMARY
research_window: [2013-01-01, 2023-12-31]
final_holdout: [2024-01-01, 2026-08-01]
stock_model: S1_sector_z_delta_short_ratio
seed_percentiles: [0.10, 0.90]
etf_model: E1_coverage_adjusted_weighted_exposure
minimum_signal_coverage: 0.50
minimum_mapping_weight: 0.80
minimum_etf_age_days: 252
minimum_median_dollar_volume: 5000000
execution: next_regular_session_open
holding_period: next_release
secondary_horizons: [5, 20, 40, 60]
transaction_cost_bps_per_side: 10
squeeze_filter: diagnostic_only
inverse_etfs: false

```

12.7 Repository structure

```
alphas/short_interest_etf/  
  data_contracts.py  
  availability.py  
  stock_features.py  
  seed_selection.py  
  reverse_index.py  
  etf_aggregation.py  
  squeeze.py  
  portfolio.py  
  diagnostics.py  
  
scripts/  
  collect_short_interest.py  
  build_release_calendar.py  
  build_short_features.py  
  build_stock_to_etf_index.py  
  run_short_interest_experiment.py  
  
tests/  
  test_short_interest_pit.py  
  test_float_pit.py  
  test_release_execution_lag.py  
  test_etf_holdings_pit.py  
  test_security_mapping.py  
  test_survivorship.py  
  
research/short_interest_etf/  
  preregistration/  
  configs/  
  results/  
  reports/
```

Chapter 13

Required Outputs and Research Report

13.1 Standard tables

13.1.1 ETF quintile table

Portfolio	Next release	5d	20d	40d	60d
Q1 high pressure					
Q2					
Q3					
Q4					
Q5 low pressure/covering					
Q5-Q1					

13.1.2 IC table

Horizon	Mean IC	Median IC	ICIR	Positive periods	N
Next release					
5d					
20d					
40d					
60d					

13.1.3 Portfolio comparison

Metric	Baseline	Avoid top pressure	Long low pressure	Long-short
CAGR				
Volatility				
Sharpe				
Sortino				
Max drawdown				
Expected shortfall				
Turnover				
Net alpha				

13.2 Required plots

Every major experiment should generate:

1. stock-level future return by short-pressure decile;
2. ETF future return by signal quintile/decile;
3. cumulative Q5–Q1 spread;
4. IC and rolling IC through time;
5. year-by-year OOS spread;
6. sector attribution;
7. breadth versus magnitude heatmap;
8. signal concentration versus return;
9. high-squeeze versus low-squeeze outcomes;
10. coverage distribution and return by coverage bucket;
11. ETF overlap cluster contribution;
12. underweight portfolio drawdown versus baseline;
13. turnover and net return versus cost assumption;
14. signal decay by forward horizon.

13.3 Experiment report template

Each report must contain:

1. exact frozen hypothesis and experiment ID;
2. Git commit and data snapshot hashes;
3. sample dates, security counts, ETF counts, and event counts;
4. PIT and survivorship audit result;
5. primary metrics and confidence intervals;
6. robustness and placebo results;
7. economic interpretation;
8. failure modes and data-quality warnings;
9. disposition: PASS, FAIL, INCONCLUSIVE, or INVALID-DATA;
10. permitted next research action.

Chapter 14

Multi-Strategy Portfolio Integration

14.1 Independent information channel

The signal should retain its own normalized output:

$$z_{j,t}^{SI} = Z_{OOS} (S_{j,t}^{ETF}), \quad (14.1)$$

where normalization parameters are estimated from training data only.

Other engines may produce:

$$\mathbf{z}_{j,t} = [z^{Analyst}, z^{Insider}, z^{Earnings}, z^{Momentum}, z^{SI}, z^{Regime}]'. \quad (14.2)$$

14.2 Do not double-count disguised signals

Estimate rolling OOS correlations and marginal predictive regressions. If short interest is simply restating negative momentum or analyst downgrades, its allocator weight should shrink.

A simple constrained combination is:

$$Combined_{j,t} = \sum_{m=1}^M \omega_{m,t} z_{j,t}^{(m)}, \quad (14.3)$$

subject to:

$$\omega_{m,t} \geq 0, \quad (14.4)$$

$$\sum_m \omega_{m,t} = 1, \quad (14.5)$$

$$\omega_{m,t} \leq \omega_{max}, \quad (14.6)$$

with weights learned only from prior OOS-like windows or fixed by pre-registration.

14.3 Confirmation/disagreement matrix

	Short interest falling/covering	Short interest rising
Analyst signal bullish	Bullish confirmation	Disagreement / crowding warning
Analyst signal bearish	Disagreement / possible capitulation	Bearish confirmation
Insiders buying	Potential informed support	Strong disagreement; investigate thesis and hedging
Momentum negative	Possible recovery/covering	Bearish confirmation

Disagreement is not necessarily a no-trade state. It can be a separate predictive feature, but only after enough observations exist.

14.4 Hedge-layer use

The central hedge engine should consume the *net portfolio exposure* after all strategies are combined. Short-interest scores can reduce the risk budget of concentrated sectors before the hedge engine decides whether index or sector hedges are needed.

Chapter 15

Risk Register and Failure Modes

Risk	How it biases results	Mitigation
Settlement-date leakage	Trades days before public disclosure	Release timestamp and next-open execution tests.
Current float backfill	Distorts historical SI ratios	PIT float; shares-outstanding fallback; versioned denominators.
Daily short-volume confusion	Treats flow as open positions	Exclude from V1; separate dataset and hypothesis.
Survivorship bias	Removes failed firms/ETFs that shorts targeted	Retain delistings, liquidations, and delisting returns.
ETF holdings look-ahead	Uses constituents not yet published	Holdings availability time and conservative lag.
Duplicate ETF exposure	Inflates independent sample size	Holdings clustering and cluster-adjusted inference.
Squeeze tail risk	Bearish signal loses violently despite eventual thesis	Separate squeeze score; ETF diversification; caps; stress tests.
Low-frequency stale data	Signal decays between reports	Natural release-cycle holding; decay analysis; optional modeled data later.
Borrow/hedging ambiguity	SI reflects non-directional positions	Event exclusions, structural flags, lending-data enhancements.
Small-cap/liquidity loading	Apparent alpha is compensation or implementation fiction	Controls, liquidity filters, value weighting, realistic costs.
Vendor revisions	Same code yields different history	Immutable raw snapshots and hashes.
Multiple testing	Random winner promoted to alpha	Pre-registration, FDR, untouched holdout.
Regime dependence	Works only in bear markets	Regime splits and persistent OOS gate.
Inverse ETF decay	Correct bearish thesis still loses through path dependency	1x first, short horizons, daily-reset simulation.

Chapter 16

Milestone Plan

16.1 Milestone A: trustworthy data

- Collect a 50-stock pilot covering size, sector, delistings, ticker changes, and high-short-interest names.
- Build release-time and denominator audits.
- Compare selected records against FINRA/exchange sources.
- Produce a data GO/NO-GO report before scaling.

16.2 Milestone B: stock signal validation

- Freeze S0/S1 and covering hypotheses.
- Run stock-level sorts and regressions.
- Evaluate squeeze, size, liquidity, and earnings splits.
- Stop if S1 has no stable OOS information.

16.3 Milestone C: ETF aggregation

- Build reverse index and point-in-time candidate universe.
- Freeze E1 and simple hit-count baseline.
- Run overlap-adjusted ETF tests.
- Add breadth/concentration only after E1 results are archived.

16.4 Milestone D: allocation and integration

- Implement underweight and long-low-pressure sleeves.
- Measure incremental value inside the multi-strategy allocator.
- Calibrate risk-budget use with walk-forward data only.
- Keep inverse and squeeze strategies isolated until their own gates pass.

Chapter 17

Appendix A: Data Dictionary

Field	Type	Timing key	Notes
security_id	string	valid-date mapping	Immutable internal ID.
settlement_date	date	effective	Regulatory position snapshot.
release_timestamp	datetime	available	First permitted information time.
short_shares	float	settlement/release	Reported open short position.
previous_short_shares	float	prior cycle	Independently validate linkage.
float_shares_pit	float	source availability	Do not backfill current float.
shares_outstanding_pit	float	source availability	Fallback denominator.
short_ratio_float	float	derived	<i>SI/Float.</i>
short_ratio_shares	float	derived	<i>SI/SharesOutstanding.</i>
delta_short_ratio	float	derived	Current minus prior ratio.
log_growth_short_ratio	float	derived	Log ratio growth with epsilon.
days_to_cover	float	settlement/release	Retain vendor definition and recomputation.
sector_z_delta	float	derived PIT	Robust sector z-score.
history_z_delta	float	derived PIT	Uses only prior release cycles.
squeeze_flag	integer	derived PIT	Separate risk feature.
etf_weight	float	holdings availability	PIT constituent weight.
signal_coverage	float	derived	Observed ETF weight.
weighted_breadth	float	derived	Weight above seed threshold.
signal_hhi	float	derived	Concentration of bearish contributions.
etf_short_score	float	derived	Exact model/version required.
experiment_id	string	immutable	Connects configuration, data, code, and outputs.

Chapter 18

Appendix B: Experiment IDs

SI-R0-DATA-PILOT
SI-R0-PIT-AUDIT
SI-R1-S0-LEVEL
SI-R1-S1-DELTA
SI-R1-S2-HISTORICAL-SURPRISE
SI-R1-COVERING
SI-R1-SQUEEZE-SPLIT
SI-R2-E1-WEIGHTED
SI-R2-COUNT-BASELINE
SI-R2-SECTOR-AGGREGATE-BASELINE
SI-R3-BREADTH-WEIGHT
SI-R3-BREADTH-COUNT
SI-R3-CONCENTRATION
SI-R4-MOMENTUM-CONTROL
SI-R4-EARNINGS-EXCLUSION
SI-R4-LIQUIDITY-SIZE
SI-R4-ETF-OVERLAP
SI-R5-UNDERWEIGHT
SI-R5-LONG-LOW-PRESSURE
SI-R5-RISK-BUDGET
SI-R6-SHORT-ETF
SI-R6-INVERSE-1X
SI-R6-INVERSE-2X
SI-R7-MULTI-SIGNAL

Chapter 19

Appendix C: Interpretation Guide

Observed result	Likely interpretation	Next action
Stock S1 works, ETF E1 does not	Information is idiosyncratic or diluted in ETF baskets	Consider stock sleeve; do not force ETF expression.
ETF E1 works better than stock	Common component is captured and idiosyncratic noise diversified	Advance to breadth and allocation tests.
Only covering side works	Low/declining short pressure is positive information	Build long-only low-pressure strategy; do not short high pressure.
Only bear-market periods work	Regime-dependent risk signal	Use as conditional risk budget, not unconditional alpha.
Relative spread works, absolute high-pressure returns positive	Correct ranking but not outright bearish timing	Underweight/avoid; reject inverse phase.
Effect disappears after momentum	Short interest repackages price action	Treat as redundant or investigate interaction only.
High breadth works, concentrated score fails	Cluster information is key; single-stock squeeze risk matters	Require breadth/effective-seed threshold.
Results vanish after ETF overlap control	Sample-size illusion from duplicate funds	Reject or redefine independent units.
Strong in-sample, weak OOS	Overfit/data-mined	Stop or redesign with a new pre-registration.

Conclusion

The most defensible initial strategy is not “high short interest means buy an inverse ETF.” It is a point-in-time research program asking whether *new, unusual short-position growth across multiple ETF constituents* predicts relative ETF weakness. The stock-first reverse-index design preserves the source information, while ETF aggregation may reduce single-name squeeze and idiosyncratic risk.

The research should advance through a strict sequence:

data validity → stock signal → ETF aggregation
→ underweight allocation → bearish implementation (19.1)
→ multi-engine integration.

A useful outcome does not require profitable inverse positions. A stable ability to reduce exposure to deteriorating ETF clusters, improve downside capture, or identify relative laggards can be valuable inside the broader portfolio and hedge architecture.

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